

PRITHVEX Calibration Report

Conformal Prediction & Uncertainty Quantification · Team Envirex

1. Why Calibration Matters

A forecast that says “demand will be 1,200 units” is far less useful than one that says “demand will be between 1,050 and 1,350 units, and we are 90% confident of that range, and historically our 90% intervals have actually contained the true value 90% of the time.” The last clause is the part most forecasting systems skip. PRITHVEX treats calibration — the property that a model's stated confidence matches its actual reliability — as a first-class requirement, not an afterthought.

2. Conformal Prediction Approach

PRITHVEX uses MAPIE-based split conformal prediction to wrap the raw ensemble forecast in an interval with a distribution-free, mathematically provable coverage guarantee. Unlike Bayesian credible intervals, which depend on the correctness of a chosen prior, conformal intervals only assume that future data is exchangeable with past calibration data — a much weaker and more defensible assumption in a live supply chain setting. This means the validity of every published interval can be checked empirically and is not simply asserted by the model.

3. Calibration Pipeline

Step	Description
1. Hold-out calibration set	A portion of historical data is withheld from training and reserved purely for calibration.
2. Conformity scores	Residuals between predictions and realised outcomes are computed on the hold-out set.
3. Interval construction	Conformity scores are used to set per-node interval widths that target the desired coverage level (e.g. 90%).
4. ECE evaluation	Expected Calibration Error compares predicted confidence to actual outcome frequency.
5. Recalibration trigger	If ECE exceeds a defined threshold, the model is automatically queued for retuning on new data.

4. Reliability Outcomes

Per the uncertainty report workflow, every forecast first passes through the Bayesian inference engine running on live supply chain data, which produces confidence-bounded forecasts per supplier, route, and warehouse node. The ECE metric is then evaluated against actual outcomes, which routes the model down one of two paths:

- Low ECE — the model is judged reliable and its confidence is treated as well-calibrated.
- High ECE — the model is automatically retuned on new data before being trusted again.

Both paths converge on a reliability diagram that is rendered as a fully auditable, explainable output — the explicit design goal being explainable calibrated intelligence with no black-box scoring.

5. Distributional Shift Monitoring

A calibrated model can still drift if the underlying data distribution changes faster than the calibration window accounts for — for example during a COVID-type disruption or a sudden geopolitical conflict. PRITHVEX runs an adaptive KL-divergence monitor in parallel with the calibration loop specifically to catch this case. When a significant shift is detected, recalibration is triggered automatically rather than waiting for the next scheduled retraining cycle.

6. Headline Calibration Results

- Approximately 30% improvement in scoring accuracy attributed to the calibration step relative to the uncalibrated baseline.
- Mathematically provable coverage guarantees on every published confidence interval, independently verifiable rather than asserted.
- Automatic, adaptive recalibration under distributional shift, with no manual intervention required during black-swan events.
- Confidence intervals delivered alongside every 7-day, 30-day, and 90-day forecast horizon.

7. Why This Matters for Decision-Makers

Calibrated uncertainty changes how a forecast can be used operationally. A planner can size safety stock to a calibrated worst-case bound rather than a single point estimate, which is the mechanism behind the platform's reported reduction in stockout frequency and the avoidance of unnecessary overstock. In short, calibration converts a probabilistic output into something a risk-averse organisation can actually plan against.